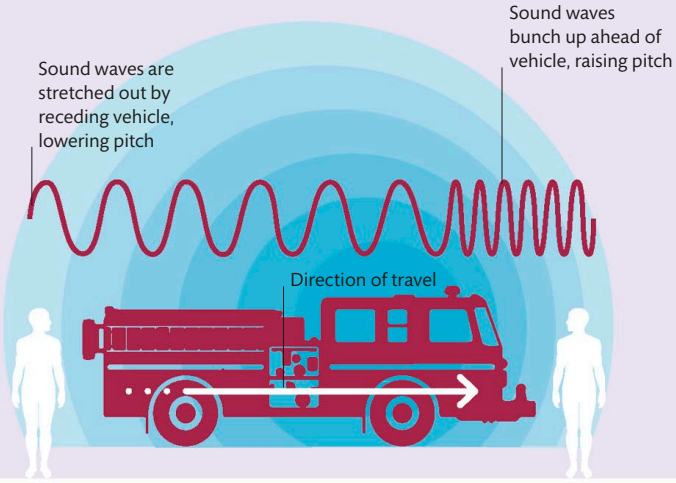
Speed cameras

A REVIEW OF 35 INTERNATIONAL STUDIES FOUND THAT SPEED CAMERAS REDUCE AVERAGE SPEEDS BY UP TO 15 PERCENT

Many types of speed cameras use radar (see pp.48–49) to measure the speed of a vehicle. They transmit radio waves at a vehicle and use the waves reflected back to calculate its speed.

The Doppler effect

When radio waves strike a vehicle that is moving toward or away from a transmitter, such as a speed camera, the vehicle's motion changes the wavelength of the reflected waves. This change is called the Doppler effect. The same effect makes an emergency vehicle's siren rise in pitch as the vehicle approaches and fall in pitch as it moves farther away.

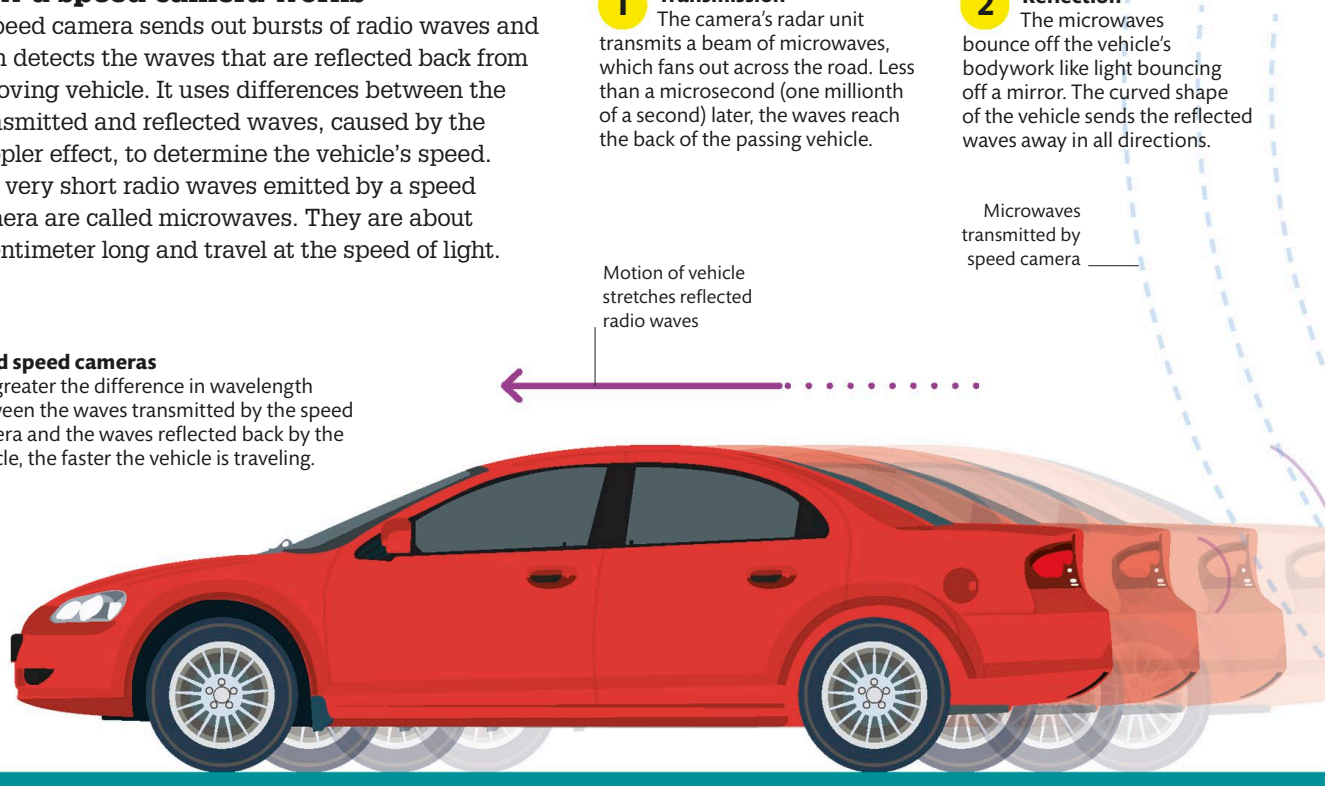


How a speed camera works

A speed camera sends out bursts of radio waves and then detects the waves that are reflected back from a moving vehicle. It uses differences between the transmitted and reflected waves, caused by the Doppler effect, to determine the vehicle's speed. The very short radio waves emitted by a speed camera are called microwaves. They are about a centimeter long and travel at the speed of light.

1 Transmission
The camera's radar unit transmits a beam of microwaves, which fans out across the road. Less than a microsecond (one millionth of a second) later, the waves reach the back of the passing vehicle.

2 Reflection
The microwaves bounce off the vehicle's bodywork like light bouncing off a mirror. The curved shape of the vehicle sends the reflected waves away in all directions.



Fixed speed cameras
The greater the difference in wavelength between the waves transmitted by the speed camera and the waves reflected back by the vehicle, the faster the vehicle is traveling.